MedVisionQuery: An Agentic Multimodal System for Medical Visual Question Answering with Explanation

Andrew Chua Kwang Hai
kwanghaiandrew_chua@mymail.sutd.edu.sg

Lai Siew Wen
siewwen_lai@mymail.sutd.edu.sg

Sai Chin Kai (Clarence)
chinkai_sai@mymail.sutd.edu.sg

Shannon Lee Yueh Ern
yuehernshannon_lee@mymail.sutd.edu.sg

Abstract

Medical visual question answering (VQA) over radiology images combines perception, question interpretation, answer synthesis and verification—steps that fail in different ways and are hidden inside a single model call. We present MedVisionQuery, an agentic system in which a Haystack Agent controller decides, case by case, how to look at the image, answer, verify the answer *against the image*, and whether to re-examine a specific region, revise, abstain or finish. Perception is question-aware, the answerer and the verifier both see the image, and a different, larger model verifies. We evaluate on 300 randomly sampled VQA-RAD test questions against a single-prompt baseline, a hand-designed workflow and a fixed-pipeline ablation, all on the same answering model, scoring with a strict rule-based scorer and a blinded LLM judge. The agentic loop is the most accurate system on the same answering model: 70.3% vs. 65.7% strict and 73.3% vs. 67.3% judged for the single prompt (McNemar $p = 0.070$ strict, $p = 0.015$ judged), ahead of the fixed pipeline (+3.0 points, $p = 0.23$) and the workflow (32.3%). It does not beat its larger verifier model answering alone (73.0%), so part of its lead may come from that model. It is more reliable—verification and self-consistency signals predict errors, it can abstain, and 8% of its answers change across re-runs versus 13% for the single prompt—at about 59× the cost. Code is available at `https://github.com/clarence-ck/med-vqa-rad-agentic`.

1 Problem Definition and Motivation

Medical VQA is a fundamentally multi-stage reasoning problem [1]. As established in the literature, a complete medical VQA system must address at minimum an image encoder, a question encoder, a multimodal fusion component, and an answer prediction module [1]—but these four components conceal distinct reasoning sub-problems that interact in clinically important ways. Given a radiology image and a clinical question, a system must separately solve: (1) *perception*—what is visible in the image that bears on the question; (2) *question interpretation*—what type of reasoning and answer is required, since different question types demand fundamentally different skills [2, 3]; (3) *synthesis*—how to integrate visual evidence into a grounded answer; (4) *verification*—whether the answer is actually supported by the visual evidence, a stage explicitly modelled in recent work as a "Confirm" module for cross-modal validation [5]; and (5) *control*—deciding, from what verification finds, whether to re-examine the image, revise the answer, abstain or stop.

These sub-tasks interact in non-trivial ways that make single-prompt approaches insufficient: what to look at depends on the question; the verifier must check cross-modal consistency between the image and the generated answer; and clinical safety requires explicit guardrails, abstention and disclaimers when confidence is low.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

Existing work: classification approaches. Early and still-prevalent medical VQA methods frame the task as multi-class classification over a fixed answer vocabulary [1]. The VQA-Med challenge [6] established the dominant paradigm: an image encoder and a question encoder produce fused features passed to a classification head. MEVF [7] extended this with meta-learning (MAML) and a denoising autoencoder (CDAE), feeding image features into a bilinear attention network (BAN) followed by a classifier. PMC-VQA [8] proposed MedVInT-TE, similarly mapping image–question pairs to answers via an encoder and classification head. Classification approaches are computationally efficient on close-ended questions but constrained to answers seen during training and perform poorly on open-ended questions requiring free-form responses.

Existing work: generation approaches. To handle open-ended questions, a parallel line of work adopts autoregressive generation. Bazi et al. [9] proposed a full transformer encoder–decoder architecture demonstrating gains on VQA-RAD and PathVQA. PMC-VQA's MedVInT-TD [8] further demonstrated that an LLM decoder can match or exceed classification-based performance when fine-tuned on large biomedical corpora. LLaVA-Med [10], accepted at NeurIPS 2023, adapted the general-domain LLaVA architecture to biomedicine via curriculum learning on PMC-15M figure–caption pairs and GPT-4-generated instruction-following data. RadFM [11] scaled this into a generalist radiology foundation model capable of VQA, report generation, and rationale diagnosis. R-LLaVA [12] extended LLaVA-Med by injecting clinician-annotated regions of interest into the CLIP encoder, improving spatial and contextual comprehension.

Existing work: hybrid approaches. Q2ATransformer [2] proposed a unified transformer framework introducing learnable candidate answer embeddings as a decoder, querying the fused image–question representation for each answer class. Although framed as classification, candidate answer embeddings interact through transformer attention, incorporating aspects of generative reasoning. This hybrid achieves state-of-the-art performance on VQA-RAD and PathVQA.

The gap that remains. Despite this progress, *all* prior approaches—whether classification [6–8], generation [9–12], or hybrid [2]—treat the task as a single end-to-end model invocation, making it impossible to localise failures, adapt the reasoning process to what the image shows, or enforce clinical safety constraints. Even recent works that acknowledge the need for verification [5] or knowledge grounding [4] address these as training objectives within a single model, rather than as explicit, auditable steps. MedVisionQuery fills this gap with an agent controller that coordinates question-aware perception, answering and image-grounded verification, deciding per case whether to re-examine the image, revise, abstain or finish, with structured failure handling.

We target the **VQA-RAD** benchmark [13], a clinically generated radiology VQA dataset; its Hugging Face release contains 2,244 image–question pairs (1,793 train / 451 test) across close-ended and open-ended question types. We evaluate on a seeded random sample of 300 test questions; no training data is used.

2 Agent Architecture

Controller loop. A fresh Haystack `Agent` is created per request (Figure 1). Its five tools are bound to per-request working memory holding the image, the question analysis, and every look, answer and check; the controller sees only short text results and chooses the next call. Unlike our earlier fixed pipeline, the tool sequence depends on what verification finds: on the 300 evaluation cases the controller finished directly in 181 cases, re-examined a region in 67, revised its answer in 34 and abstained in 18. Budgets cap the loop at two focused re-looks, three answers and 16 steps, with a safety net that checks the latest answer if the controller stalls.

Roles. The *controller* is the planner: it chooses every next step from the results so far. The per-request *working memory* (image, question analysis, looks, answers, checks) is the memory; the tools read from and write to it. `analyze_question` (task understanding) classifies the question type and expected answer format. `look(focus)` (perception) examines the image *with the question in mind*; with a focus, it zooms: the named region is located, cropped and enlarged, and examined beside the full image; the first look rejects non-radiology images. `answer(guidance)` (executor) answers from the image plus all observations, applying the verifier's guidance on revisions. `check` (evaluator) first reads the image *blind*—the verifier answers from the image and question alone, without the

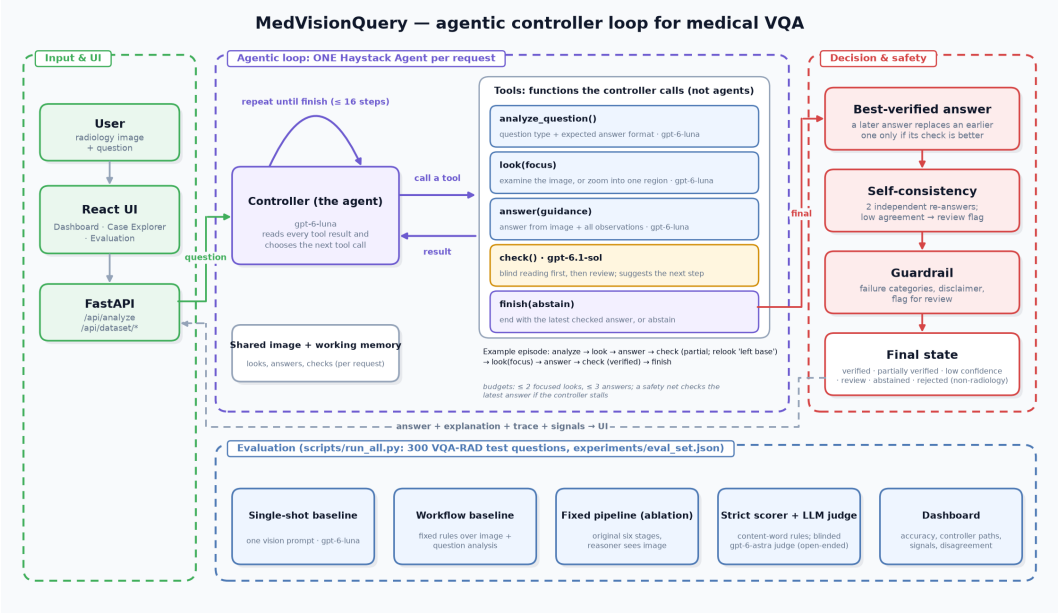


Figure 1: MedVisionQuery. One Haystack Agent (the controller) calls five tools in a loop, reading each result before choosing the next: question analysis, question-aware perception (`look`), answer synthesis, image-grounded verification (`check`) and `finish`. The tools are not agents; `check` only recommends a re-look, a revised answer, abstention or acceptance, and the controller decides. A decision-and-safety stage keeps the best-verified answer, measures self-consistency and applies guardrails.

proposed answer or the looker's notes—then verifies the answer against the image with five checks; an answer is only verified if the blind reading agrees. It returns a status (verified, partially verified, low confidence, unsupported) and recommends *accept*, *relook* with a focus, *reanswer* with guidance, or *abstain*. `finish(abstain)` ends the episode and is refused while the latest answer is unchecked.

Model and tool choices. All roles use OpenAI models, sized by need and configurable per role. The controller and question analysis make simple decisions (`gpt-6-luna`, no/low reasoning effort); perception and answering use `gpt-6-luna` with medium effort, and the answerer always sees the image, because a text description loses the visual detail many questions depend on. Verification uses a different, larger model (`gpt-6.1-sol`) so that the answering model does not verify itself. Haystack's `Agent`/`Tool` abstraction provides the tool-calling loop and exit conditions; keeping the image in per-request memory rather than in the controller's context keeps the controller cheap. An earlier retrieval stage over SLAKE [15] was removed: it returned other patients' question–answer pairs, which contradicted yes/no and location answers and gave no measurable benefit on perception-bound questions.

Baselines and ablation. The *single-shot baseline* sends the image and question in one prompt. The *workflow baseline* is a fixed chain—a question-agnostic image description, question analysis, then hand-written rules per question type. The *fixed-pipeline ablation* runs our original six stages (interpret, analyse, filter, route, reason, verify, with retry) in a fixed order with the image given to the reasoner, isolating what the controller's decisions add. A demo UI (Section 6) exposes all systems.

3 Multimodal Grounding

The system uses two modalities, the radiology image and the text question, and they interact at every step that determines the answer: the question (and any focus) steers what `look` examines; `answer` and `check` receive the image directly alongside the text observations; only the controller works purely in text. Our earlier design described the image once, *without* the question, and passed only that text downstream, so many answers came back as "cannot be determined from the image

Table 1: Accuracy (%) on 300 random VQA-RAD test questions (same answering model for all systems). Open-ended accuracy is strict/judged. Time is per-case latency in seconds (mean/median), measured with four cases running concurrently; the whole run of all four systems took 58 minutes of wall-clock time. Cost is the mean API cost per case.

System	Strict	Judged	Close (169)	Open (131)	Time (s)	Cost
Single-shot baseline	65.7	67.3	77.5	50.4 / 54.2	3.6 / 2.8	\$0.0002
Workflow baseline	32.3	34.7	46.7	13.7 / 19.1	7.2 / 6.8	\$0.0005
Agentic loop	**70.3**	**73.3**	81.1	56.5 / 63.4	42.4 / 27.6	\$0.0139
Fixed pipeline (ablation)	67.3	69.3	76.9	55.0 / 59.5	30.2 / 24.0	\$0.0067

description"; every model that answers or judges now sees the image itself. Because the verifier sees the image, its "explanation grounded in evidence" and "no contradictions" checks are genuine cross-modal agreement checks.

4 Control and Failure Handling

Detection. Image-grounded verification (five checks and a status), seven failure categories (*unsupported claim*, *low confidence*, *visual contradiction*, *partially verified*, *question–answer mismatch*, *generation unstable*, *parser failure*) and an eighth, *self-inconsistent*, from resampling. **Recovery.** Instead of re-running the whole pipeline, the controller zooms into the region the verifier names or revises the answer with the verifier's guidance. When the blind reading disagrees, the controller is told only *that* it disagrees, never the blind answer, so the loop must resolve it with its own model. A later answer replaces an earlier one only if its check is strictly better, because focused re-looks can over-read. **Self-critique.** After finishing, the answerer is re-asked independently twice; low agreement adds a failure category and flags the case for review. **Fallback.** The controller may abstain (reported as an abstention, never coerced into "no"); a guardrail adds a disclaimer and a review flag for low-confidence, unsupported, high-severity or self-inconsistent cases; non-radiology or unreadable images are rejected before any answer is produced.

5 Evaluation

Setup. 300 VQA-RAD test questions drawn at random, keeping the split's close/open-ended mix (seed 42; 169 close-ended, 131 open-ended); all four systems run on the same cases with the same answering model. *Strict scoring*: yes/no answers must start with the correct word; binary terms must name the correct term and not its opposite; counts must match; open-ended answers must contain every content word of the reference (aliases and plurals folded; $\geq$60% for references longer than three words), with left/right conflicts and denials counted wrong and embedding similarity [14] only as a strict fallback. Because word matching both rejects correct paraphrases and accepts vague answers that contain the reference word, a *blinded LLM judge* (`gpt-6-astra`, a model that neither answers nor verifies in any system; answers shuffled and anonymised) scores every open-ended answer as a second, independent measure; a 30-verdict spot check agreed with 28 credit decisions (the other two lenient, on workflow answers). Paired comparisons use the exact McNemar test; 95% intervals are about $\pm$5.2 points. The baseline and agentic loop were re-run on 100 cases to measure stability. API cost of the evaluation: \$8.81 (\$6.40 runs, \$1.31 repeats, \$1.10 judge). Wall-clock time: 58 min for the main run (four cases at a time), 68 min for the sequential repeats and 16 min for the judge.

Results. Table 1. The agentic loop leads the single prompt (33 vs. 19 discordant cases, $p = 0.070$); under the judge the lead is significant (34 vs. 16, $p = 0.015$). Against the same models in a fixed order it leads but not significantly (27 vs. 18, $p = 0.23$; judged $p = 0.081$), and both far exceed the hand-designed workflow. Gains concentrate where the controller acts: on re-examined cases the loop scored 49% versus 43% for the single prompt, and on revised cases 71% versus 47% (Table 2). The verifier model answering alone—the check's blind reading—scores 73.0% strict ($p = 0.33$ against the loop), so the loop does not exceed, and numerically trails, the larger model it consults.

Table 2: Left: accuracy (strict, %) by the path the controller took. Right: agentic accuracy by its own signals.

Path	n	Agentic	Single	Pipeline	Signal	Agentic
direct	181	85	80	83	verified (205)	84
relook	67	49	43	45	partially verified (87)	45
reanswer	34	71	47	47	unsupported/low (8)	0
abstained	18	–	39	33	consistency high (248)	77
					consistency med/low (34)	55–65

Failure case analysis. *Self-consistency:* two independent re-answers per case; high consistency means 77% accuracy, medium/low 55–65%, so low consistency marks uncertainty rather than certain error. *Cross-modal agreement:* because the verifier sees the image, verification status is predictive: verified answers are 84% correct, unsupported or low-confidence answers 0%. *Verifier model:* a larger model verifying a smaller one steers 119 of 300 episodes away from the direct path. *Disagreement:* the three systems disagree on 68% of cases; when exactly one system is right, it is the agentic loop 27 times, the single prompt 14 and the workflow 14. *Stability:* on 100 repeated cases, 13% of single-prompt answers flipped between right and wrong, versus 8% for the agentic loop. Four patterns dominate the errors: (i) a shared perception blind spot, where looker and verifier misread the image the same way; (ii) over-reading on re-looks, limited by keeping the best-verified answer; (iii) benchmark items whose question does not match the image, which every system fails; and (iv) workflow rigidity, where fixed rules answer "Brain" or "X-ray" when a specific finding is needed.

6 User Interface

A React frontend exposes three views: **Dashboard** (upload an image, compare the agentic loop with both baselines, inspect the trace, verification checks and failure categories), **Case Explorer** (browse the 300evaluation cases with full agent traces) and **Evaluation** (strict and judged accuracy for all four systems, accuracy by question type and by controller path, disagreement and a per-case table).

7 Reflection

When agentic behaviour is beneficial. The loop is the most accurate system built on the same answering model, and its gains sit exactly where the controller intervenes (revised and re-examined cases). The accuracy gain over a single prompt is modest and not significant on every score, and the loop does not beat its larger verifier answering alone, so we attribute part of it to that model rather than to control alone. Its clearer benefit is *trustworthiness*: verification and consistency signals identify likely errors, it abstains rather than guess, its answers change less between runs, and every decision is auditable—valuable in clinical review, at a substantial cost.

Limitations. About 59× the cost and 10× the median latency of a single prompt for a modest accuracy gain. The verifier shares the perception weaknesses of vision models, so agreement is not correctness. 300 questions give ±5.2-point intervals and noisy per-type estimates; the LLM judge is itself a model, spot-checked but not human-annotated. A retrieval stage did not help perception-bound questions and was removed; retrieval may still help questions that need knowledge not visible in the image, which VQA-RAD rarely contains.

References

[1] Lin, Z., Zhang, D., Tao, Q., Shi, D., Haffari, G., Wu, Q., He, M., & Ge, Z. (2023). Medical visual question answering: A survey. *Artificial Intelligence in Medicine*, 143, 102611.

[2] Liu, Y., Wang, Z., Xu, D., & Zhou, L. (2023). Q2ATransformer: Improving medical VQA via an answer querying decoder. In *Information Processing in Medical Imaging – IPMI 2023*, Lecture Notes in Computer Science, vol. 13939, pp. 445–456. Springer.

[3] Liu, B., Zhan, L.M., Xu, L., & Wu, X.M. (2023). Medical visual question answering via conditional reasoning and contrastive learning. *IEEE Transactions on Medical Imaging*, 42(5), 1532–1545.

[4] Qiu, C., Huang, K., Xie, Z., Liu, M., Gu, J., & Zong, X. (2025). Explainable medical visual question answering via chain of evidence. *Knowledge-Based Systems*. doi:10.1016/j.knosys.2025.113672.

[5] Syed, A., et al. (2025). MedVCTP: Improving accuracy and explainability in medical visual reasoning. In *NeurIPS 2025 Workshop on Human-AI Coevolution (HAIC)*.

[6] Ben Abacha, A., Hasan, S.A., Datla, V.V., Liu, J., Demner-Fushman, D., & Müller, H. (2019). VQA-Med: Overview of the medical visual question answering task at ImageCLEF 2019. *Working Notes of CLEF 2019*.

[7] Nguyen, B.D., Do, T.T., Nguyen, B.X., Do, T., Tjiputra, E., & Tran, Q.D. (2019). Overcoming data limitation in medical visual question answering. In *MICCAI 2019*, Lecture Notes in Computer Science, pp. 522–530. Springer.

[8] Zhang, X., Wu, C., Zhao, Z., Lin, W., Zhang, Y., Wang, Y., & Xie, W. (2023). PMC-VQA: Visual instruction tuning for medical visual question answering. *arXiv preprint arXiv:2305.10415*.

[9] Bazi, Y., Al Rahhal, M.M., Bashmal, L., & Zuair, M. (2023). Vision–language model for visual question answering in medical imagery. *Bioengineering*, 10(3), 380.

[10] Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., & Gao, J. (2023). LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day. In *Advances in Neural Information Processing Systems (NeurIPS 2023), Datasets and Benchmarks Track*.

[11] Wu, C., Zhang, X., Zhang, Y., Hui, H., Wang, Y., & Xie, W. (2025). Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *Nature Communications*, 16, 7866.

[12] Chen, X., Lai, Z., Ruan, K., Chen, S., Liu, J., & Liu, Z. (2025). R-LLaVA: Improving Med-VQA understanding through visual region of interest. In *ICLR 2025 Workshop on Human-AI Coevolution (HAIC)*. `https://arxiv.org/abs/2410.20327`.

[13] Lau, J.J., Gayen, S., Ben Abacha, A., & Demner-Fushman, D. (2018). A dataset of clinically generated visual questions and answers about radiology images. *Scientific Data*, 5:180251.

[14] Deka, P., Jurek-Loughrey, A., & Deepak, P. (2022). Improved methods to aid unsupervised evidence-based fact checking for online health news. *Journal of Data Intelligence*, 3(4), 474–504. `https://huggingface.co/pritamdeka/S-PubMedBert-MS-MARCO`.

[15] Liu, B., Zhan, L.M., Xu, L., Ma, L., Yang, Y., & Wu, X.M. (2021). SLAKE: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. In *Proceedings of the IEEE 18th International Symposium on Biomedical Imaging (ISBI 2021)*, pp. 1650–1654. IEEE.